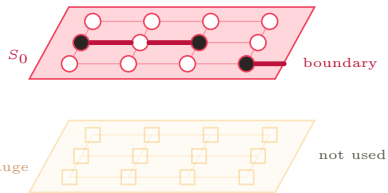
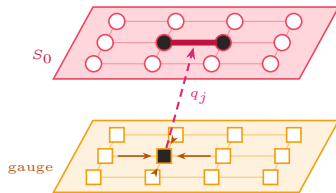
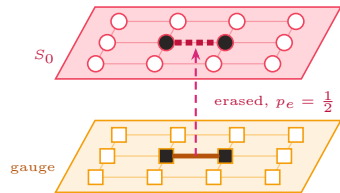


(a) MWPM

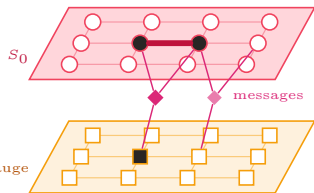
Match the S_0 detection events. Edge weights $w_e = \ln[(1 - p_e)/p_e]$.

(b) Sequential BP

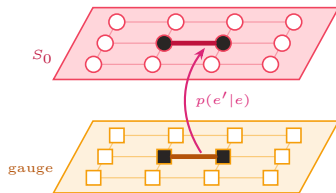
BP on the link detectors gives fault posteriors q_j , which set the S_0 edge probabilities.

(c) Erasure passing

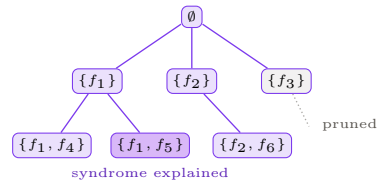
Match the link detectors first. S_0 edges of the matched faults become erasures with $p_e = \frac{1}{2}$.

(d) Belief-matching

BP on the full detector error model, including faults that touch both levels. Posteriors reweight matching.

(e) Hierarchical correlated matching

Pass 1 matches both levels. Every matched edge e lowers the weight of its partners e' to $p(e'|e)$. Pass 2 matches again.

(f) Tesseract

Best-first search over sets of faults on the full detector error model, ordered by the cost $\sum_j w_j$ plus a lower bound for the unexplained detectors.

○ S_0 detector □ gauge detector ● detection event — matched edge - - - erased edge ♦ fault (BP factor)